Digital Evidence Analysis Report: Case 001

Investigator: Drew
Case name (Autopsy): Digital_Evidence_Analysis_Case001
Date of examination: 3 November 2026 (all times UTC)
Environment: Parrot OS 7 (ARM64) virtual machine, The Sleuth Kit 4.12.1, Autopsy 2.24-6, Foremost 1.5.7, PhotoRec 7.2

1. Investigation Overview

The objective of this investigation was to demonstrate digital forensic examination techniques: file-system analysis, timeline reconstruction from timestamps and logs, deleted-file recovery by data carving and by other forensic methods, and correlation of independent artifacts into a factual sequence of events.

A controlled 256 MB ext4 disk image was created containing sample documents, a log file, and two deliberately deleted files. Because the events were created by the examiner, every finding can be checked against known ground truth. The scenario is simulated and does not establish the identity or intent of any real user.

2. Tools and Technologies

Tool	Purpose
Autopsy 2.24	Web-based forensic browser: file listing, deleted files, metadata, timeline, keyword search
The Sleuth Kit 4.12.1	`fsstat`, `fls`, `istat`, `icat`, `blkstat`, `blkcat`, `mactime` for command-line analysis
Foremost 1.5.7	Signature-based file carving
PhotoRec 7.2	Second, independent carving tool
sha256sum / md5sum	Evidence integrity verification

3. Evidence Acquisition and Preservation

The image was built with `dd if=/dev/zero of=forensic_image.dd bs=1M count=256`, formatted with `mkfs.ext4`, and mounted via loopback. Sample files were written five seconds apart so their timestamps would be distinguishable. Two files were then deleted, the image was synced and unmounted, and it was hashed.

The original image was set read-only (`chmod 444`). A working copy was made and all analysis was performed on the copy.

Item	SHA-256
forensic_image.dd (original)	d7112134ca40c4003600d932ded3dd9a480029fb40cf1987b0d38e2eba265d6a
forensic_working.dd (copy)	d7112134ca40c4003600d932ded3dd9a480029fb40cf1987b0d38e2eba265d6a

Both images also produced the MD5 `889728cb91bec76dd0dd3ffd040633d3`, which matches the MD5 Autopsy calculated when the image was imported. This shows Autopsy analyzed the same verified working copy. The hashes were recorded at 19:24:38 UTC.

Screenshot: hash verification (02_hash_verification.png)

4. File-System Analysis

`fsstat` identified an ext4 filesystem with a 1024-byte block size and 65,537 inodes. The filesystem was created and first mounted at 19:16:58 and last written at 19:23:26.

`fls -r -p` listed three directories (Documents, Logs, Downloads) and three live files. `fls -r -d -p` listed two deleted inodes, 32775 and 32776, under `$OrphanFiles`. ext4 clears the directory link on deletion, so the original filenames and parent folder are no longer recoverable from metadata. The inode numbers match the numbers recorded by `find -ls` before deletion, which ties the orphan entries to the original files.

Inode	Original file	Size before deletion	State at analysis
32772	Documents/investigation.txt	44 B	Allocated, block 139777
32773	Logs/access.log	56 B	Allocated, block 140289

Inode	Original file	Size before deletion	State at analysis
32774	Documents/account_changes.txt	39 B	Allocated, block 140801
32775	Downloads/deleted_evidence.txt	46 B	Not allocated, size 0
32776	Downloads/deleted_report.pdf	142 B	Not allocated, size 0

Autopsy showed the same directory listing and listed inodes 32775 and 32776 in red under All Deleted Files, with deletion times of 19:22:43 and 19:22:54.

Screenshots: fsstat, fls, fls -d, Autopsy file listing, Autopsy deleted files

5. Deleted-File Recovery

5.1 Metadata recovery (Sleuth Kit `icat`): unsuccessful

Both deleted inodes show size 0, zero links and an empty block list in `istat` and in Autopsy. `icat` returned 0-byte files for both. Autopsy reports the recovered content MD5 as `d41d8cd98f00b204e9800998ecf8427e`, the hash of an empty file. ext4 clears the extent pointers when a file is deleted, so the data blocks remain on disk but the inode no longer points to them.

5.2 Signature carving (Foremost): successful

`foremost -v -t pdf` carved one PDF of 142 bytes at byte offset 143,132,672 (block 139778). `file` identified it as PDF 1.4, and `blkstat` confirmed the block is not allocated.

5.3 Signature carving (PhotoRec): successful

PhotoRec, searching free space, recovered one file, `f0279556.pdf`, from sector 279556. This is the same location Foremost reported (279556 x 512 = 143,132,672).

5.4 Keyword recovery of the text file: successful

A plain `.txt` file has no signature to carve. `strings -t d` found "Deleted evidence recovery test - Document 001" at byte offset 144,704,512, which is block 141313. `blkstat` shows the block is not allocated. The 46 bytes were extracted with `blkcat`. Autopsy's keyword search of unallocated space independently returned one hit in fragment 141313 (status Not Allocated, group 17).

5.5 Comparison of methods

Method	Examines	Target	Result	SHA-256
icat	Inode metadata	Text and PDF	Failed, 0 bytes	n/a
Foremost	File signatures	PDF	Recovered, 142 B	54725981e65c4cb6635d6e2faa8ac15070c84f169b66d21aa669a8ceedca939c
PhotoRec	File signatures, free space	PDF	Recovered, 142 B	54725981e65c4cb6635d6e2faa8ac15070c84f169b66d21aa669a8ceedca939c
strings / blkcat / Autopsy keyword	Raw and unallocated blocks	Text	Recovered, 46 B	f4085f33e4b6a1d7982ae6e8332ef84e44321968b785ef8edeae4d76d5c2dc0a

The two carvers produced bit-identical output. As a lab-only check, a regenerated copy of the original PDF content hashed to the same value, and a freshly hashed copy of the original text string matched the recovered text. A real case would have no original to compare against.

The contrast is the main result: metadata-based recovery failed because the pointers were cleared, while content-based methods succeeded because the data still sat in unallocated space.

6. Timeline Reconstruction

The timeline was generated with `fls -r -m "/" -z UTC` and `mactime -b ... -d -y -z UTC`. The bodyfile has 10 lines (SHA-256 `b75d38323e7f990a6b353888657418e25575b1c3dec2ac9dbcc0dea5962b7503`) and the timeline has 19 lines (SHA-256 `f22d106e2c8ae58b587c0cac9ca1504dcc66018eca88a6f6a7655a62e39f3452`). Autopsy's timeline view produced the same sequence.

Time (UTC)	Event	Evidence
19:16:58	Filesystem created and first mounted	fsstat; /lost+found born

Time (UTC)	Event	Evidence
19:17:41	Documents, Logs, Downloads created	`...b` entries
19:18:09	investigation.txt created (44 B)	inode 32772 macb
19:18:37	access.log created (56 B); /Logs modified	inode 32773 macb; directory m.c.
19:19:04	account_changes.txt created (39 B); /Documents modified	inode 32774 macb; directory m.c.
19:20:25	deleted_evidence.txt created (inode 32775)	`.a.b`
19:20:44	deleted_report.pdf created (inode 32776)	`.a.b`
19:20:55	Directory access cluster, consistent with a directory traversal (the `find -ls` command)	`.a..` entries
19:22:43	Inode 32775 deleted, size set to 0	`m.c.`, Deleted time
19:22:54	Inode 32776 deleted; /Downloads modified	`m.c.`, Deleted time
19:23:05	/Downloads accessed (directory listing)	`.a..`
19:23:26	Filesystem last written	fsstat
19:24:38	Images hashed and original locked read-only	examiner record

The sequence matches the commands the examiner ran. Directory timestamps corroborate file events independently: /Logs changed at the moment access.log was created, and /Downloads changed at the moment the last file was removed.

7. Artifact Correlation and Findings

1. **Inode continuity.** The inode numbers recorded before deletion (32775, 32776) match the orphaned inodes found afterwards, and the carved PDF size (142 B) matches the pre-deletion size.
2. **Independent tool agreement.** Sleuth Kit and Autopsy produced matching timelines. Foremost and PhotoRec produced identical PDFs from the same offset.
3. **Log content versus file metadata.** access.log contains the entry `2026-11-03 10:15:00`, but the file was created at 19:18:37 UTC, roughly nine hours later. Content written inside a file is controlled by whoever writes it, so it cannot be relied on alone.

Filesystem metadata is a more independent source. The discrepancy is itself a finding and would need explanation in a real case (for example, a different time zone, a clock difference, or a staged entry). Here the entry was written by the examiner as a sample.

4. **Recovered text and its source.** The recovered text is consistent with deleted_evidence.txt: it has the same length (46 B) and content, and sits in the block that continues the allocation pattern of the earlier files (139777, 140289, 140801, 141313, each 512 blocks apart). The filename and inode link were lost on deletion, so this is circumstantial and the report states it as consistent with, not proof.

Timestamps and log entries establish a supported sequence of events, not who performed them.

8. Windows Registry Analysis

Registry analysis was performed on NIST's CFReDS Windows registry dataset (cfreds-2017-winreg, Windows 10 build 10586), not on the Linux image. Three scripted user actions were matched across the action log, a process event log and the registry hives, with agreement to within seconds for two of them.

8.1 Data source and preservation

The archive `cfreds-2017-winreg_sgrd-10.7z` (796 MiB) has SHA-256 `1d5ebaadb64ebf92b0234db7f27eb376dc818de263b186c8a2fda8640b88f610`. It extracted without error to 983 files in 181 folders. **Open item:** compare this hash with the value NIST lists for this download before submission.

Folder	Contents
0x01_action_log	One CSV of scripted user actions (the answer key)
0x02_event_log	467 CSVs of process-monitor style events, local time
0x03_screen_capture_by_virtualbox	Screen recordings of the session
0x04_reference_hive	Hives in `p1`, plus shadow copies `p1_vss1` to `p1_vss3`, for users CFReDS, CFTT, cfttu and Default

The extracted files were copied to a working directory and all 983 files were hashed (SHA-256 `2582b0ba528576de636cb201df2e1cfc4a40d4182d66de3f0314dd1a7bdc87d8` for the hash list). The originals were then made read-only.

8.2 Method

Hives were read with `reglookup` 1.0.1 from the working copy. The action log records Pacific Time (UTC-08:00), and registry key last-write times are stored in UTC, so log times were converted by adding eight hours. The event log shows local 12-hour times, assumed to be the same zone; its agreement with the hive timestamps supports that.

8.3 Findings

#	Scripted action (log, Pacific to UTC)	Event log (UTC)	Hive key and last-write (UTC)	Assessment
1	seq 28, `net user CFTT ... /add`, 16:58:53 = 2016-11-13 00:58:53	`lsass.exe` RegCreateKey on `SAM\SAM\DOMAINS\Account\Users\Names\CFTT` at 00:59:00.645	`/SAM/Domains/Account/Users/Names/CFTT`, 2016-11-13 00:59:00	Match to the second. 7.6 s from command to commit
2	seq 29, `net localgroup Administrators CFTT /add`, 16:59:10 = 00:59:10	`lsass.exe` RegCreateKey on `Builtin\Aliases\Members\S-1-5-21-...\000003EB` at 00:59:12.153, then RegSetValue on `Builtin\Aliases\00000220\C`	`Builtin/Aliases/Members/S-1-5-21-.../000003EB`, 2016-11-13 00:59:12	Match. 2.2 s after the command
3	seq 5 and 10, "Disable UAC", 16:54:21 and 16:54:53 = 00:54:21 and 00:54:53	None: event logs start at 16:57:47 Pacific	`Policies/System` key, 2016-11-13 00:54:43. `EnableLUA` = 1, `ConsentPromptBehaviorAdmin` = 0, `PromptOnSecureDesktop` = 0	Consistent, not exact

Finding 1. The CFTT account has RID 0x3EB (1003). The same `lsass.exe` burst created the name key, the user record `Users\000003EB` (with `F` and `V` values) and an alias membership entry, so one RID ties the three artifacts together. The event log and the hive agree, which also confirms the time zone conversion.

Finding 2. The alias key `Builtin/Aliases/00000220` has a later last-write (2016-11-14 01:56:25), so it cannot date this event; it was rewritten about a day afterwards. Its raw value contains the text "Administrators", consistent with 0x220 being the built-in

Administrators alias. `reglookup` does not decode SAM binary values, so the membership change rests on the event-log write plus the membership key's timestamp, not on a decoded group list.

Finding 3. The log says UAC was disabled, but `EnableLUA` is still 1. The two prompt values are 0, which fits prompts being turned off rather than UAC being off. The key was last written 22 s after the first command and 10 s before the second, and a key holds only its most recent write, so the match is consistent but not precise. No event-log entry covers it, because monitoring began later.

8.4 Registry limitations

- The dataset is scripted, so a ground-truth log exists. A real case would not have one.
- `reglookup` reads the main hive files only. Recent changes held in the `.LOG` transaction files were not replayed.
- SAM binary values (user records, group member lists) were not decoded.
- Other accounts in the hive (`cfttu`, `cfttu_8jsax5x`) were seen but not examined.
- Only a Windows 10 data set was examined, and only three actions were correlated.

9. Evidence Register and Chain of Custody

ID	Item	Hash	Recorded (UTC)	Handler
EV-001	forensic_image.dd (original, read-only)	SHA-256 d7112134…265d6a	19:24:38	Drew
EV-002	forensic_working.dd (working copy)	SHA-256 d7112134…265d6a	19:24:38	Drew
EV-003	Foremost PDF 00279556.pdf	SHA-256 54725981…ceedca939c	see log	Drew
EV-004	PhotoRec PDF f0279556.pdf	SHA-256 54725981…ceedca939c	see log	Drew
EV-005	recovered_evidence_carved.txt	SHA-256 f4085f33…5d2dc0a	see log	Drew
EV-006	bodyfile.txt	SHA-256 b75d3832…962b7503	see log	Drew

ID	Item	Hash	Recorded (UTC)	Handler
EV-007	timeline.csv	SHA-256 f22d106e...39f3452	see log	Drew
EV-008	Autopsy unallocated extract	MD5 C7EC67D22218311038331FBA82CED17B	see log	Drew

Fill in the exact recording times from your terminal history before submitting.

Registry evidence added to the register:

ID	Item	Hash	Recorded (UTC)	Handler
EV-009	cfreds-2017-winreg_sgrd-10.7z (NIST registry dataset)	SHA-256 1d5ebaad...b88f610	see log	Drew
EV-010	registry_hashes.txt (SHA-256 of all 983 extracted files)	SHA-256 2582b0ba...c87d8	see log	Drew
EV-011	registry_work copy of the extracted dataset	per-file hashes in EV-010	see log	Drew

10. Limitations

- The image is a controlled Linux ext4 filesystem, not a real workstation. Registry analysis uses a separate NIST reference dataset (section 8), not this image.
- Recovery depends on the deleted data blocks not being overwritten. In this small, quiet image they were not, which is unrealistically favorable.
- The text file was recovered by keyword search, not by signature, and its link to inode 32775 is circumstantial.
- The filesystem timeline begins at filesystem creation. The `dd` and `mkfs` commands are not visible in file timestamps.
- Autopsy 2.24 displayed an "Error Parsing File" message for the synthetic `$OrphanFiles` entry (inode 65537), which has no real timestamps. This is a display quirk and does not indicate image corruption.

- Autopsy 2.24's launcher uses Perl taint mode (`-wT`), which rejected the PATH on this system. A copy of the launcher was run with taint mode removed (`-w`). The installed binary was not modified.
- All content in the image was authored by the examiner. Nothing here identifies a real person.

11. Conclusion

The investigation demonstrated file-system analysis, timeline reconstruction, three-way deleted-file recovery, and evidence integrity verification. Metadata recovery failed on ext4, while Foremost, PhotoRec and keyword search all recovered deleted content from unallocated space, and the recovered files matched known originals by hash. Independent tools agreed on every result. A separate NIST registry dataset supported the registry component: scripted account creation and group changes matched the action log, the process event log and the SAM hive to within seconds.

12. Figure Index

Screenshot files to insert at the matching section. Names are the files saved on your machine.

Fig.	File (Screenshot_2026-11-03_at_...)	Shows	Section
1	2_08_42_PM	Tool installation, Autopsy package check	2
2	2_11_55_PM	Tool versions, aarch64	2
3	2_21_17_PM	Sample files created, `find -ls`	3
4	2_24_58_PM	Deletion and SHA-256 match of both images	3
5	2_26_04_PM	`fsstat` , `fls` , `fls -d`	4
6	3_24_48_PM	`istat` for inodes 32775, 32776, 32772, 32773	4, 6
7	3_17_41_PM	`icat` returning 0-byte files	5.1

Fig.	File (Screenshot_2026-11-03_at_...)	Shows	Section
8	3_19_17_PM	Foremost run and audit	5.2
9	3_35_05_PM	PhotoRec recovery summary	5.3
10	3_35_52_PM	PhotoRec file hashes	5.3
11	3_40_27_PM	`istat 32774`, reference PDF hash, `blkstat 139778`, `strings` hit	5.2 to 5.4
12	3_42_43_PM	`blkstat` / `blkcat` of block 141313 and text hash match	5.4
13	3_43_51_PM	`mactime` timeline	6
14	3_48_28_PM	Bodyfile and timeline hashes	6
15	3_56_58_PM	Autopsy new case	4
16	4_04_00_PM	Autopsy MD5 and image import	3
17	4_04_34_PM	Autopsy file listing	4
18	4_05_01_PM	Autopsy deleted files	4
19	4_07_09_PM	Autopsy timeline	6
20	4_12_45_PM	Inode 32776 metadata and MD5 match	3, 5.1
21	4_14_05_PM	Unallocated space extraction	5.4
22	4_17_01_PM	Autopsy keyword hit, fragment 141313	5.4
23	4_33_25_PM	Registry archive SHA-256	8.1
24	4_35_13_PM	Registry archive extraction	8.1
25	4_37_44_PM	Dataset folders and hive files	8.1
26	4_38_41_PM	Action log	8.3
27	4_39_56_PM	Working copy, 983 hashes	8.1
28	4_41_54_PM	System hive locations	8.1
29	4_43_23_PM	SAM: CFTT key last-write	8.3, finding 1

Fig.	File (Screenshot_2026-11-03_at_...)	Shows	Section
30	4_46_50_PM	Event log columns	8.3
31	4_48_10_PM	Event log: `lsass.exe` creates CFTT key	8.3, finding 1
32	4_49_42_PM	Alias writes and hive membership key	8.3, finding 2
33	4_45_01_PM	SOFTWARE `Policies/System` values	8.3, finding 3